Problem 10.3

Assume that the force of interest varies with time and is given by

$$\delta_t = a + \frac{b}{t}.$$

Find the formula for the accumulation of one unit money from time t_1 to time t_2.

Solution

For a time-varying force of interest, the accumulation factor from time t_1 to time t_2 is

$$\exp\left(\int_{t_1}^{t_2} \delta_t \, dt\right).$$

Here,

$$\delta_t = a + \frac{b}{t}.$$

Thus the accumulation of one unit from t_1 to t_2 is

$$\exp\left(\int_{t_1}^{t_2} \left(a + \frac{b}{t}\right) dt\right).$$

Evaluate the integral:

$$\int_{t_1}^{t_2} \left(a + \frac{b}{t}\right) dt = \left[at + b \ln t\right]_{t_1}^{t_2}.$$

Therefore,

$$\left[at + b \ln t\right]_{t_1}^{t_2} = a(t_2 - t_1) + b\big(\ln t_2 - \ln t_1\big).$$

Using logarithm rules,

$$\ln t_2 - \ln t_1 = \ln\left(\frac{t_2}{t_1}\right).$$

Hence the accumulation factor is

$$e^{a(t_2 - t_1) + b \ln(t_2/t_1)}.$$

This can also be written as

$$e^{a(t_2 - t_1)} \left(\frac{t_2}{t_1}\right)^b.$$

Thus, the accumulation of one unit from time t_1 to time t_2 is

$$\boxed{e^{a(t_2 - t_1)} \left(\frac{t_2}{t_1}\right)^b}.$$